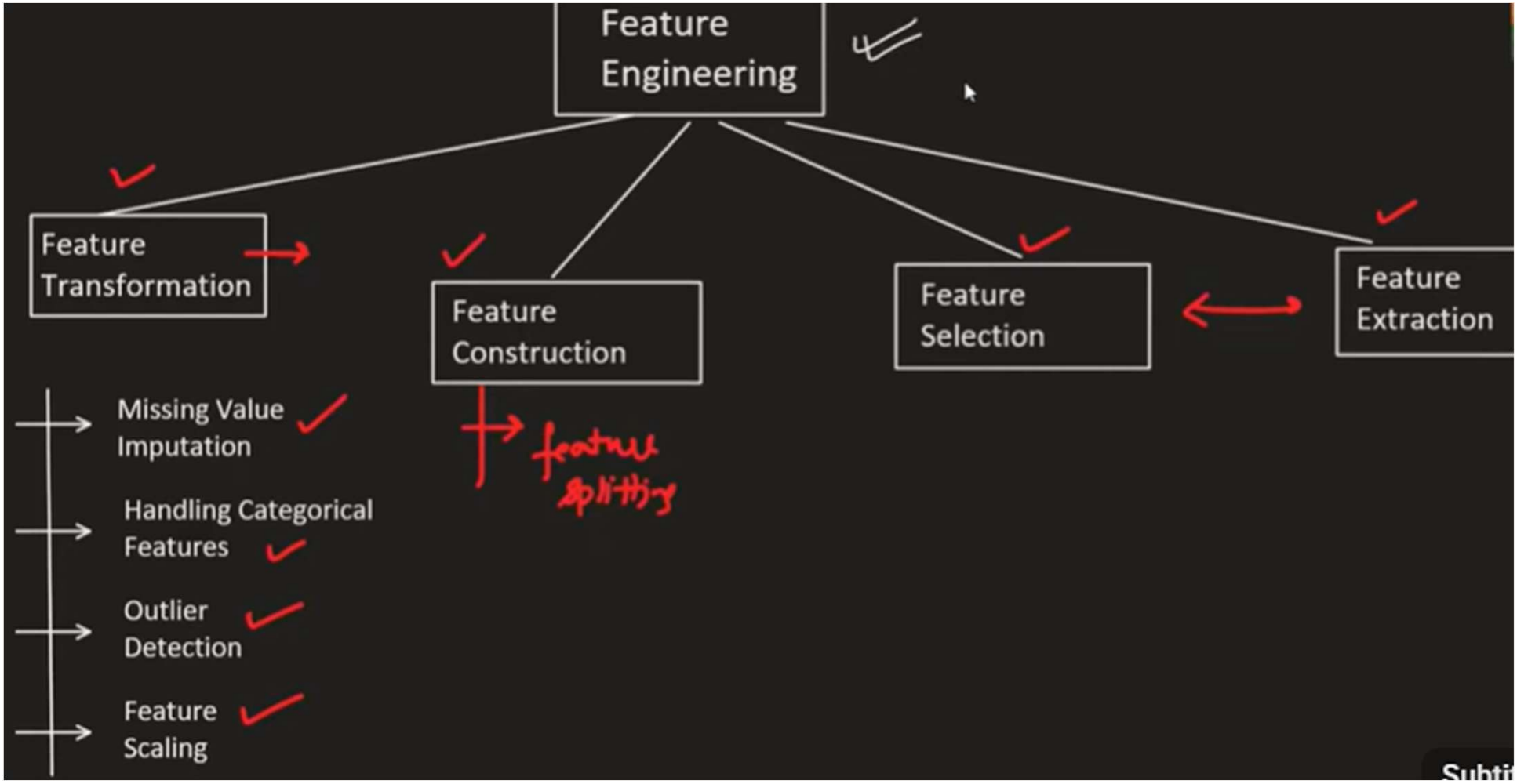
Principle Component Analysis

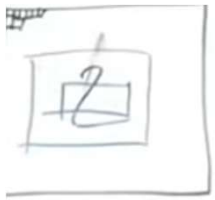
Dr. Muhammad Safyan


Feature Engineering
Feature Transformation
Feature Construction
Feature Selection
Feature Extraction
Missing Value Imputation
Handling Categorical Features
Outlier Detection
Feature Scaling

Curse of Dimensionality

- Features→ dimension
- As number of feature increase , performance of model increase.
- Optimal number of features are best for performance , other wise model will be complex.
- If we increase beyond that optimality figure, performance will decreases

- Handwritten digit classifier dataset.
- Image, text data are always high dimension

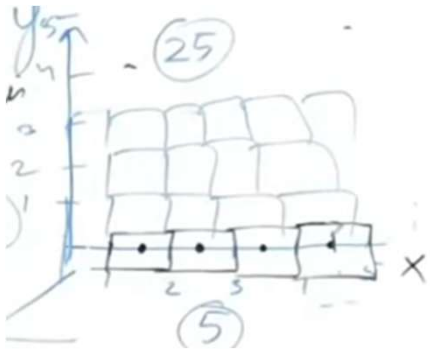


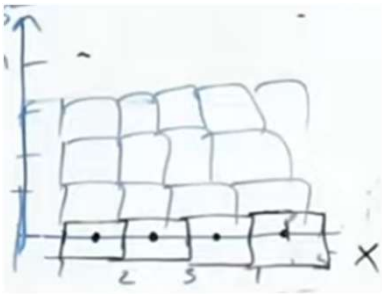
- Higher dimensionality
- Sparsity
- In statistical technique. To nearest point become far

from each other

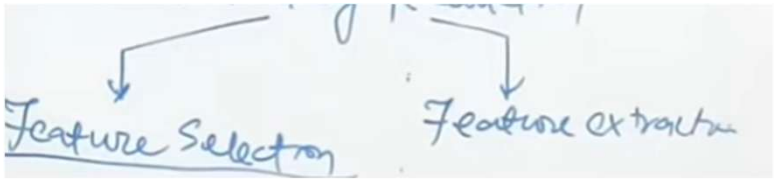
Performance Decrease

Computation increases





- **Dimensionality Reduction**



Feature Selection:

F1,f2, ……. Fn

F1, f2, f5

1)Forward Selection
2)Backward Elimination

Feature Extraction:

We make new features from the existing features (PCA)

New features are the linear combination of the existing features.

Principal Component Analysis(PCA)

- PCA is feature extraction
- Unsupervised Techniques
- Will reduce the features
- PCA convert higher dimension to lower dimension (best possible lower dimensions), while keeping the essence of data,
- 3-D to 2-D(best possible lower Dimension)

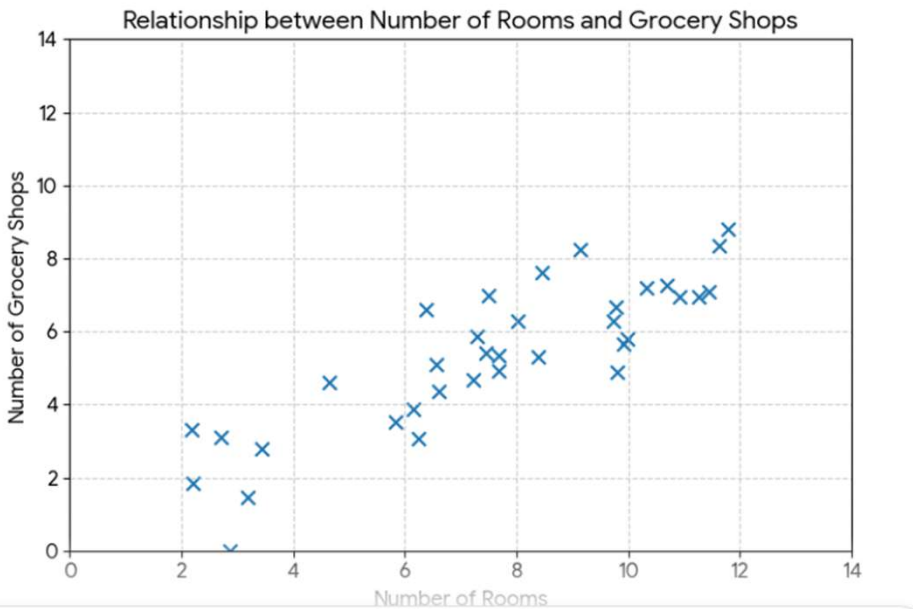
Benefits

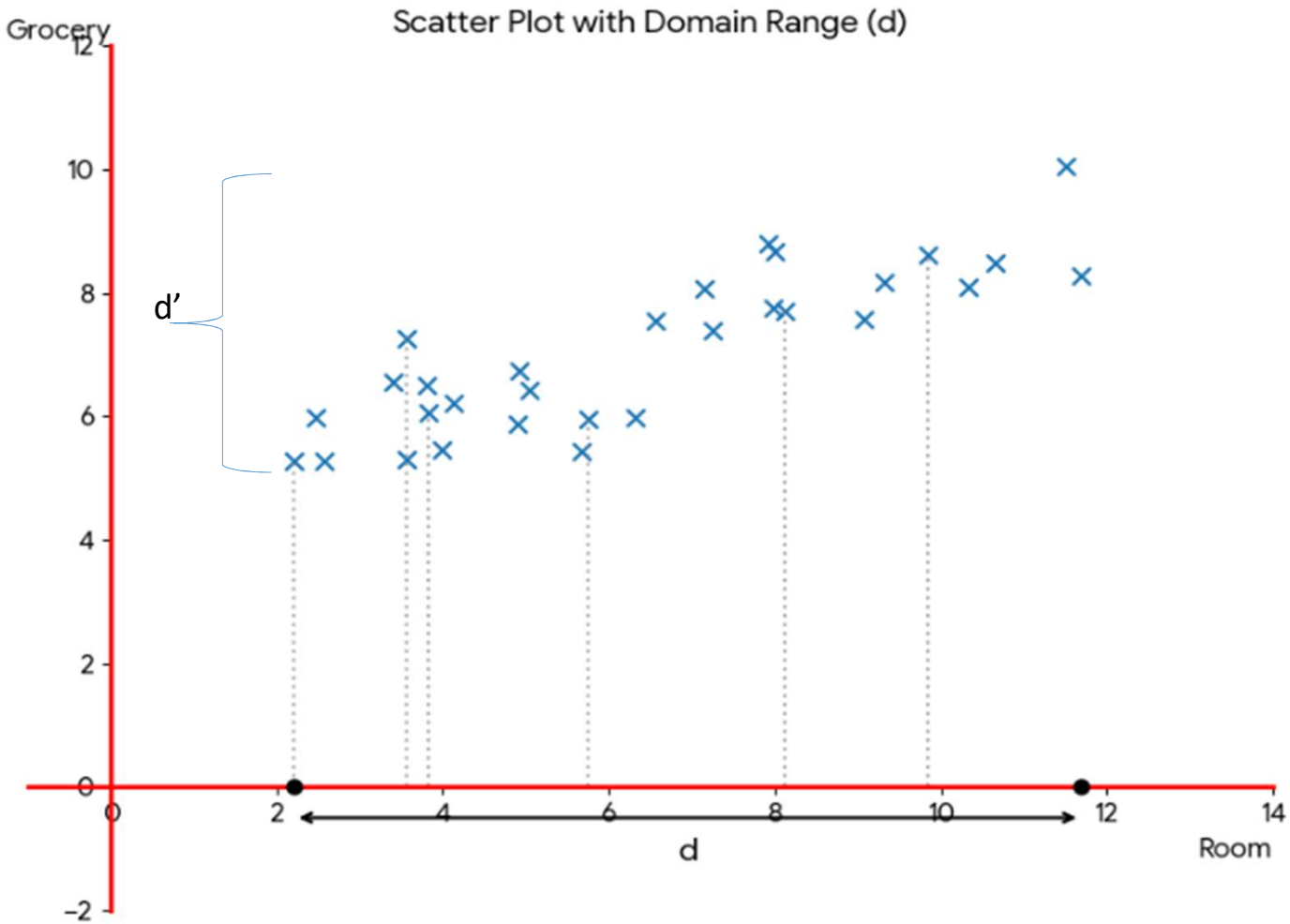
- Reduce the size of the data.
- Faster execution of algorithm(computation become save)
- Better Visualizations
  - (Minist dataset 784→ 3D).

Geometric Institutions

No. of Rooms	No. of Grocery Shops	Price (L)
3	2	60
4	0	130
5	6	170
2	10	90

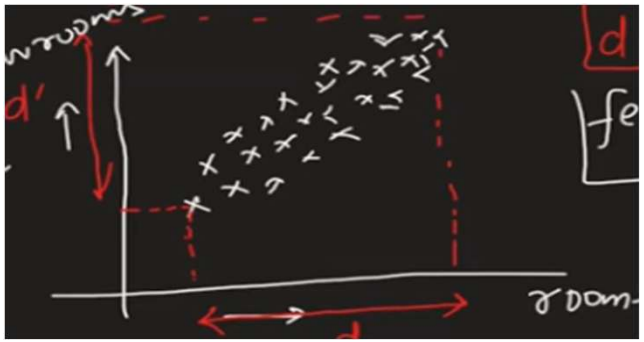
- Consider first Feature selection:
- If choose one→ you will select No. of Rooms
- If you don't have domain knowledge , and how you have to select few feature from n features.
- You need a trick for feature selection


Relationship between Number of Rooms and Grocery Shops
Number of Grocery Shops
Number of Rooms
0
2
4
6
8
10
12
14


Scatter Plot with Domain Range (d)
Grocery
12
10
8
6
4
2
0
-2
d'
0
2
4
6
8
10
12
14
d
Room

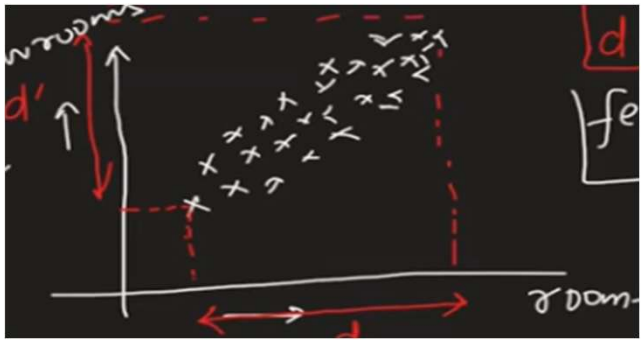
d>d'

No. of Rooms	No. of Washrooms	Price



No. of Rooms	No. of Washrooms	Price

Size	Price

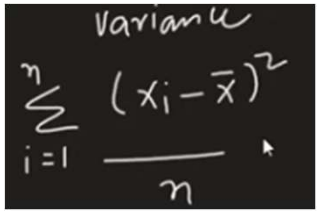


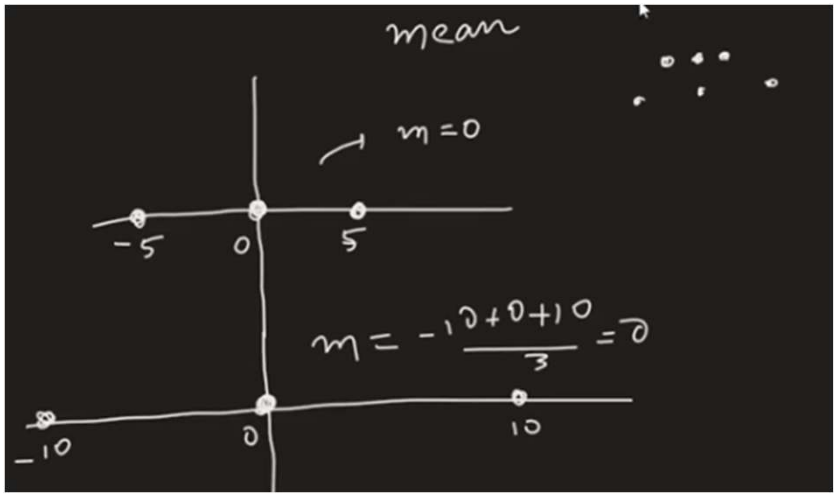
- You will get the to principle compoentsPc1, pc2
- You will choose the , whose variance is more i.e pc1
- And you will transform the data according to pc1
- Pc<= n

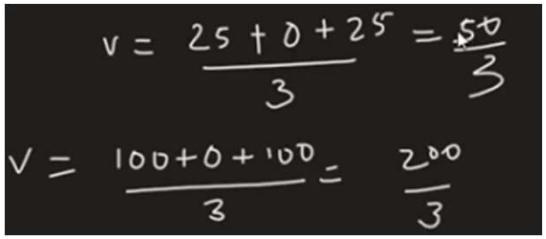
Why the variance is so important

- **Mean:**
- Mean tells the central tendency

- Variance is the Spread of data

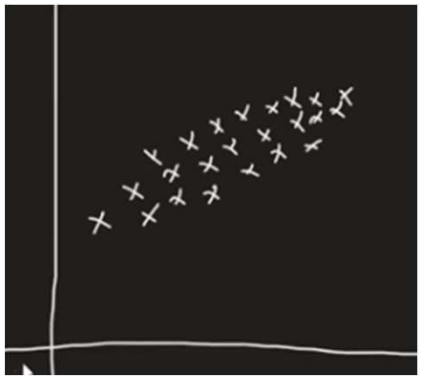


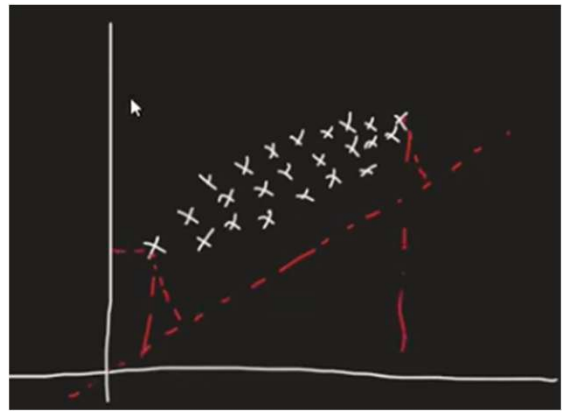


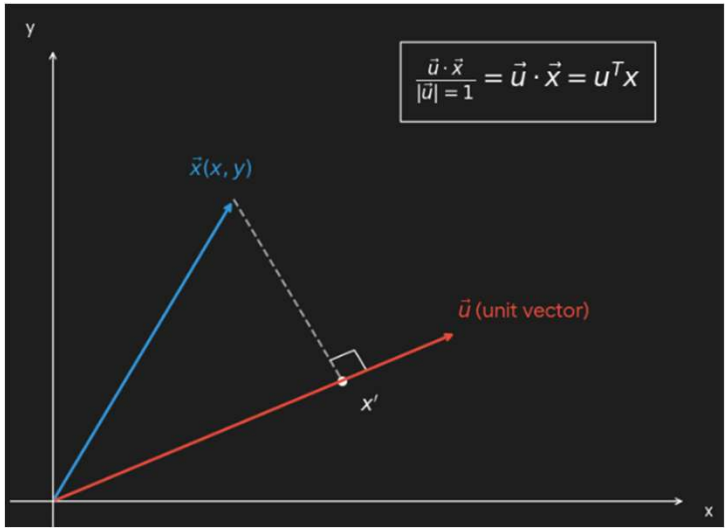


Variance importance

PCA







$$\frac{u \cdot x}{|u| =}$$

$$\frac{u \cdot x}{|u| = 1}$$

$$u \cdot x$$

$$u^T x$$

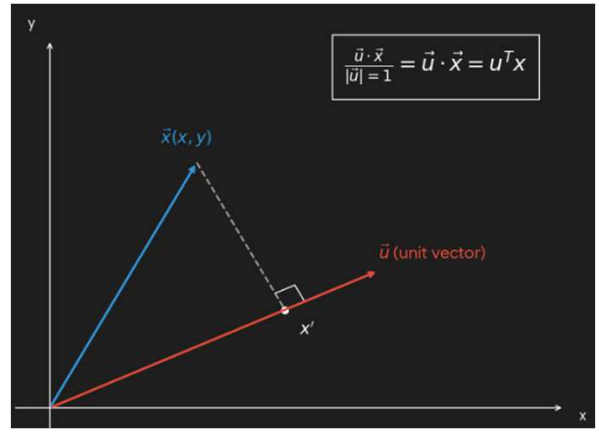
Will produce a scalar

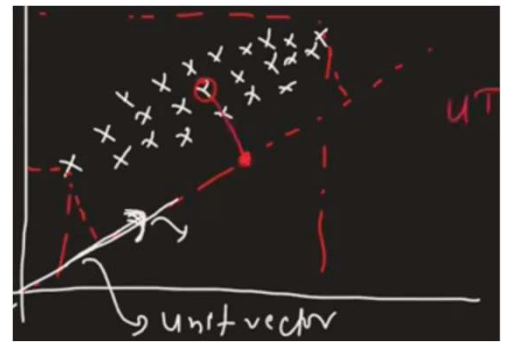
- Goal is to maximize the variance
- There can be Uni-vector anywhere
- Which uni-vector will be suitable.
  - Whose variance will be maximum

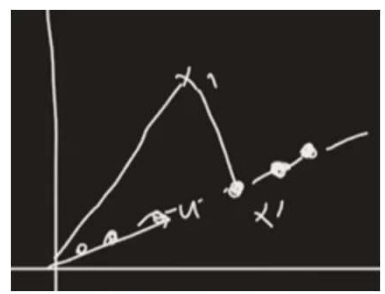
$$\frac{1}{n}\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2$$

$$\sum_{i=1}^{n}\left(u^T x_i - u^T \bar{x}\right)^2$$

Become an Optimization problem

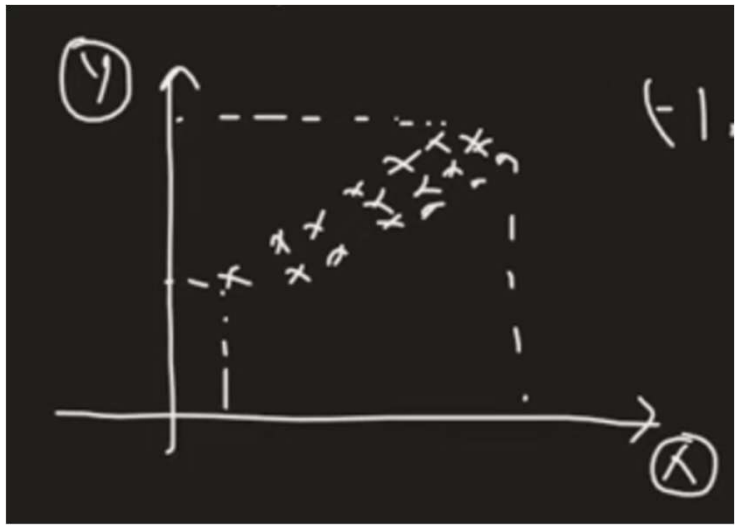


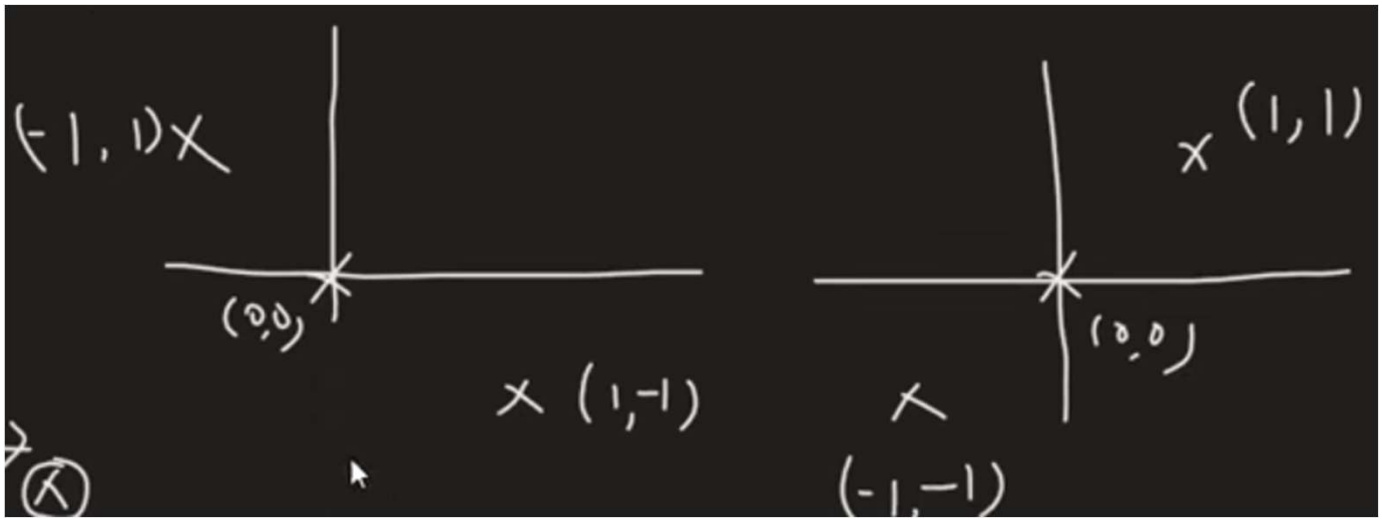




Covariance and Covariance Matrix

- Mean is the central tendency, does not tell the spread
- Variance tells about the spread
- Variance will give a number but does not tell the relationship among the multi-dimensions





- Variance will same. But they are different
- Covariance tell the relationship tell the relationship among the dimensions

$$\frac{-1+0-1}{3}$$

$$-\frac{2}{3}$$

$$\frac{1+0+1}{3}$$

$$\frac{2}{3}$$

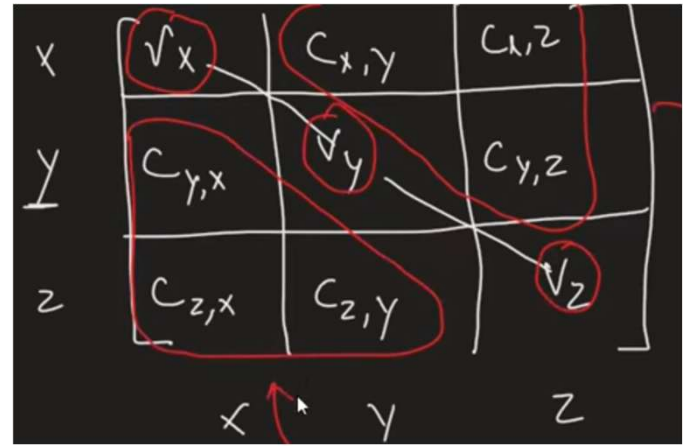
- Correlation is similar to covariance
- You bound the correlation between -1 and 1

Covariance Matrix

$$\begin{bmatrix} cov(x_1, x_1) & cov(x_1, x_2) \\ cov(x_1, x_2) & cov(x_2, x_2) \end{bmatrix}$$

$$\begin{bmatrix} Var(X_1) & Var(X_1, X_2) \\ Cov(X_2, X_1) & Var(X_2) \end{bmatrix}$$

x_2



Note that Var(X1,X2)*Var*(*X*1,*X*2) and Cov(X2,X1)*Cov*(*X*2,*X*1) are actually the same thing (covariance is symmetric), so this matrix is symmetric

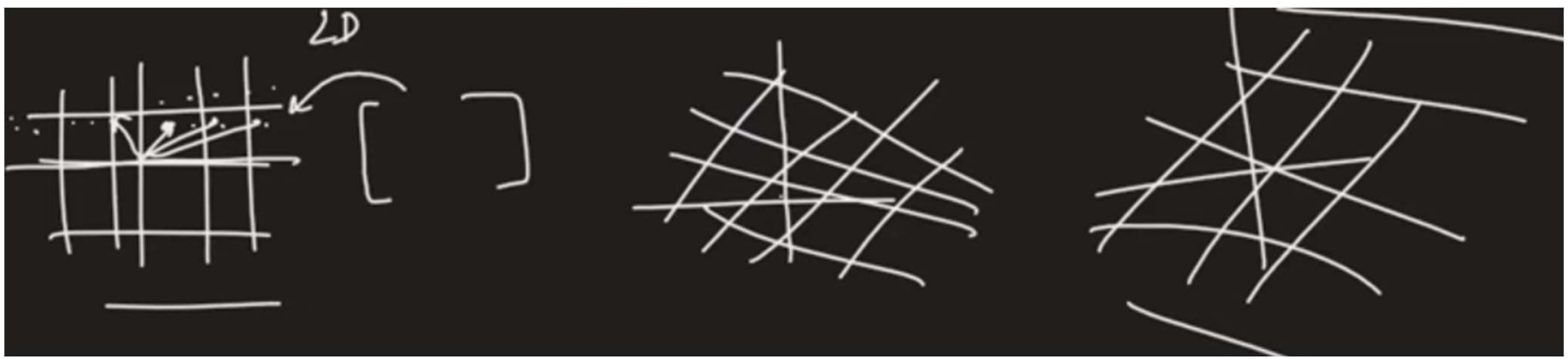
cov(a,b)=cov(b,a)

Every axis of covariance tell the variance and relation between the axis's.

Covariance has tell two things:
spread on each axis
And relationship between the axis

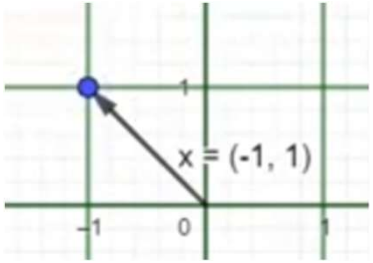
Linear Transformation, Eigen vector & Eigen value

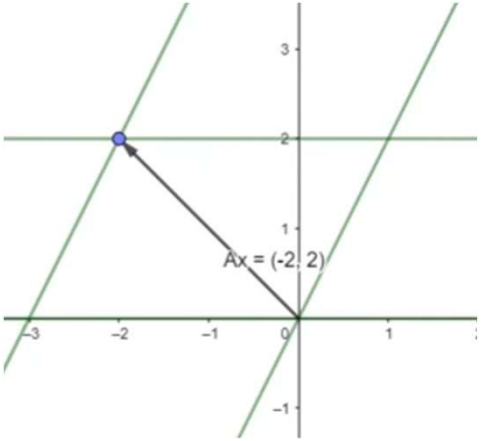
- Matrices as Linear Transformation:
- In coordinate system, there are infinite data point, each data point is a vector . When apply matrices on the coordinate system, it will transform the vectors.



Eigen Vector

- Are special vector. When apply transformation. Whose direction will not change. Magnitude may change.
- You always vectors(two eigen vectors)

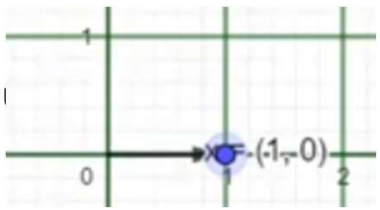


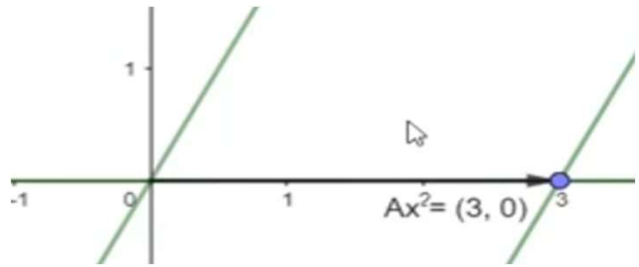


Eigen values

- how much vector stretch or shrink after transformation is

- Eigen val





$$\vec{A} \cdot \vec{V} = \lambda \vec{V}$$

$$\sum_{i=1}^{n} \left(u^T x_i - u^T \bar{x}\right)^2$$

- For Matrix A, use covariance matrix
- Covariance matrix define both spread and orientation.
- If a transformation has largest eigen value, that direction will have maximum variance.
- Process is called Eigen Decomposition

Step by step solution of PCA

F1	F2	F3	traget

Step-1: Mean center in each dimension

- Step-2 find covariance matrix

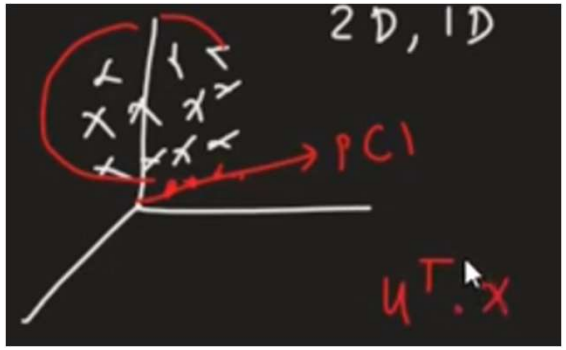
$$
\begin{array}{c}
f_1 \\ f_2 \\ f_3 \\ \\
\end{array}
\begin{bmatrix}
V(f_1) & C(f_1, f_2) & C(f_1, f_3) \\
C(f_2, f_1) & V(f_2) & C(f_2, f_3) \\
C(f_3, f_1) & C(f_3, f_2) & V(f_3)
\end{bmatrix}
$$

$$
\begin{array}{ccc}
f_1 & f_2 & f_3
\end{array}
$$

- Step-3- find the eigen value and eigen vector for covirance matrix
- If you are with 3-D data you will find 3 vector and 2 lambdas: lambd1, lambda2, lambda3.

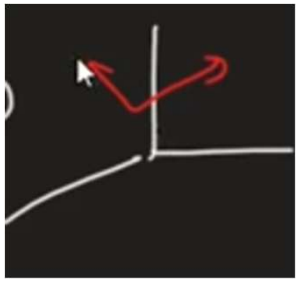
- Lambda with greatest value is pc-1, gradually decreasing will be pc-2, pc-3.
- It is up to you , you want to go with 2-d or 1-d.

F1	F2	F3	traget



- Pc-1 dimension(1,3)
- Data dimension(1000,3)
- Resultant will be =1000,1

- For 2-d



- Pc-1,pc-2 dimsnion will be (2,3)
- Data dimension(1000,3)
- Resultant will be =1000,2

Pc-1	Pc-2	target

PCA using Covariance Matrix (Eigen Method)

1. Mean Center the Data
    1. Xcentered=X−μ
2. Compute Covariance Matrix $C = \frac{1}{n} X^T X$

3. Compute Eigenvalues Cv=λv
4. Compute Eigen Vector
5. Normaize egine vector
6. Sort Eigenvalues (Descending)
7. Select Top k Eigenvectors
8. Project Data
    1. Xreduced=X·Vk

Sample Dataset

.et's take 2D data (very common in exams):

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 3 & 1 \\ 1 & 3 \end{bmatrix}$$

Step 1: Mean Center Data

Mean of columns:

- Mean of col1 = $(2+0+3+1)/4 = 1.5$
- Mean of col2 = $(0+2+1+3)/4 = 1.5$

Centered data:

$$X_c = \begin{bmatrix} 0.5 & -1.5 \\ -1.5 & 0.5 \\ 1.5 & -0.5 \\ -0.5 & 1.5 \end{bmatrix}$$

- Step 2: Covariance Matrix

$$C = \frac{1}{4} X_c^T X_c$$

$$C = \begin{bmatrix} 1.25 & -0.75 \\ -0.75 & 1.25 \end{bmatrix}$$

- Step 3: Eigenvalues & Eigenvectors
- Use Eigen Equation

$$Cv = \lambda v$$

Convert to determinant form:

$$\det(C - \lambda I) = 0$$

- Next step : Form C−λI

$$C - \lambda I = \begin{bmatrix} 1.25 - \lambda & -0.75 \\ -0.75 & 1.25 - \lambda \end{bmatrix}$$

- Next Step: Compute Determinant

$$\begin{vmatrix} 1.25-\lambda & -0.75 \\ -0.75 & 1.25-\lambda \end{vmatrix} = 0$$

$$(1.25-\lambda)^2 - (-0.75)^2 = 0$$

$$(1.25-\lambda)^2 - 0.5625 = 0$$

- Next Step: Find the Eigen Values

$$(1.25-\lambda)^2 = 0.5625$$

$$1.25-\lambda = \pm 0.75$$

Case 1:

$$1.25-\lambda = 0.75 \Rightarrow \lambda = 0.5$$

Case 2:

$$1.25-\lambda = -0.75 \Rightarrow \lambda = 2$$

- Eigenvalues

$$\lambda_1 = 2, \quad \lambda_2 = 0.5$$

- Find Eigenvector for λ=2

$$(C - 2I)v = 0$$

$$\begin{bmatrix} -0.75 & -0.75 \\ -0.75 & -0.75 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

Equation:

$$-0.75x - 0.75y = 0$$

$$x + y = 0 \Rightarrow x = -y$$

Eigenvector:

$$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Find Eigenvector for $\lambda = 0.5$

$$(C - 0.5I)v = 0$$

$$\begin{bmatrix} 0.75 & -0.75 \\ -0.75 & 0.75 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

Equation:

$$0.75x - 0.75y = 0$$

$$x = y$$

Eigenvector:

$$v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- Step 7: Normalize Eigenvectors (Important)

Normalize $v_1 = [1, -1]$

$$||v_1|| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Normalize $v_2 = [1, 1]$

$$||v_2|| = \sqrt{2}$$

$$v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Final Answer

Eigenvalues:

$$\lambda_1 = 2, \quad \lambda_2 = 0.5$$

Eigenvectors (unit):

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- Step 5: Project Data
  - Xreduced=Xc·v1

$$X_c = \begin{bmatrix} 0.5 & -1.5 \\ -1.5 & 0.5 \\ 1.5 & -0.5 \\ -0.5 & 1.5 \end{bmatrix} \qquad v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

PCA using SVD (Preferred in Big Data)

- Instead of covariance matrix, we directly apply SVD on data.
- Mean Center the Data
  - Xcentered=X−μ
- Apply SVD

$$X = U\Sigma V^T$$

- Extract Principal Components
  - Columns of V = principal directions
- Singular Values → Variance
  - Relationship:
  - Singular values give eigenvalues

$$\lambda_i = \frac{\sigma_i^2}{n}$$

- Select Top k Components
  - Choose top columns of V
- Project Data
  - $X_{reduced} = X \cdot V_k$

PCA using SVD

- Step 1: Mean Center Data

$$X = \begin{bmatrix} 0.5 & -1.5 \\ -1.5 & 0.5 \\ 1.5 & -0.5 \\ -0.5 & 1.5 \end{bmatrix} \quad (4 \times 2)$$

🔵 Step 1: Understand Goal

We want:

$$X = U\Sigma V^T$$

Where:

- U: 4×2 (orthonormal columns)
- Σ: 2×2 (singular values)
- V: 2×2 (orthonormal)

Step 2: Compute $X^T X$

$$X^T = \begin{bmatrix} 0.5 & -1.5 & 1.5 & -0.5 \\ -1.5 & 0.5 & -0.5 & 1.5 \end{bmatrix}$$

Algorithms like:

- Golub–Kahan SVD
- QR-based methods

Directly compute:

$$X = U\Sigma V^T$$

$$X^T X = \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix}$$

🔵 Step 3: Eigenvalues (for singular values)

We already know for:

$$\begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix}$$

$$\det(X^T X - \lambda I) = 0$$

$$\begin{vmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{vmatrix} = 0$$

$$(5-\lambda)^2 - 9 = 0$$

$$\lambda = 8,\ 2$$

👉 Eigenvalues:

$$\lambda_1 = 8, \quad \lambda_2 = 2$$

Step 4: Singular Values

$$\sigma_1 = \sqrt{8} = 2\sqrt{2}, \quad \sigma_2 = \sqrt{2}$$

$$\Sigma = \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}$$

ght Singular Vectors (V)

$$v_1 = \frac{1}{\sqrt{2}}[1, -1], \quad v_2 = \frac{1}{\sqrt{2}}[1, 1]$$

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

For $\lambda_1 = 8 \rightarrow$ find v_1

Plug into $(X^TX - 8I)v = 0$:

$[[5-8, -3], [-3, 5-8]] = [[-3, -3], [-3, -3]]$

Row 1: $-3v_1 - 3v_2 = 0$

$\rightarrow v_1 = -v_2$

Set $v_2 = -1 \rightarrow v_1 = 1$. Raw vector: $[1, -1]^T$

Normalize: $\|[1,-1]\| = \sqrt{(1^2+(-1)^2)} = \sqrt{2}$

$v_1 = (1/\sqrt{2})[1, -1]^T$

Check: both rows give same equation → consistent ✓

For $\lambda_2 =$

Plug into $(X^TX - 2I)v = 0$:

$[[5-2, -3], [-3, 5-2]] = [[3, -3],$

Row 1: $3v_1 - 3v_2 = 0$

$\rightarrow v_1 = v_2$

Set $v_1 = v_2 = 1$. Raw vector: $[1, 1]$

Normalize: $\|[1,1]\| = \sqrt{(1^2+1^2)} = \sqrt{}$

$v_2 = (1/\sqrt{2})[1, 1]^T$

Orthogonality check: $v_1 \cdot v_2 = (1)$

$V = [v_1 \mid v_2] = (1/\sqrt{2})[[1, 1], [-1, 1]]$ (columns are the eigenvecto

PC1 direction $= v_1 = [1/\sqrt{2}, -1/\sqrt{2}]^T$ PC2 direction $= v_2 = [1/\sqrt{2}, 1/$

Step 6: Compute $U = \frac{Xv}{\sigma}$

- **Compute u_1**

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Compute Xv_1:

$$Xv_1 = \begin{bmatrix} 0.5 & -1.5 \\ -1.5 & 0.5 \\ 1.5 & -0.5 \\ -0.5 & 1.5 \end{bmatrix} \begin{bmatrix} 0.7071 \\ -0.7071 \end{bmatrix} = \begin{bmatrix} 1.4142 \\ -1.4142 \\ 1.4142 \\ -1.4142 \end{bmatrix}$$

Divide by $\sigma_1 = 2\sqrt{2} = 2.828$:

$$u_1 = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ -0.5 \end{bmatrix}$$

- **Compute u_2**

$$v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$Xv_2 = \begin{bmatrix} -0.7071 \\ -0.7071 \\ 0.7071 \\ 0.7071 \end{bmatrix}$$

Divide by $\sigma_2 = \sqrt{2} = 1.414$:

$$u_2 = \begin{bmatrix} -0.5 \\ -0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

Step 7: Final U Matrix

$$U = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

Final SVD Answer

Get Plus ×

$$X = U\Sigma V^T$$

U

$$\begin{bmatrix} 0.5 & -0.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

Σ

$$\begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix}$$

V

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

SVD → PCA projection

🔵 Step 1: What PCA does

👉 PCA projects data onto **principal directions (columns of V)**

$$Z = XV$$

- Z = transformed data (principal components)
- Columns of Z = PC1, PC2

🔵 Step 2: Take first principal component (PC1)

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

🔹 Project data onto PC1

$$Z_1 = Xv_1$$

Compute row by row:

↓

✅ PC1 values

$$Z_1 = \begin{bmatrix} 1.4142 \\ -1.4142 \\ 1.4142 \\ -1.4142 \end{bmatrix}$$

🔵 Step 3: Second principal component (PC2)

$$v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- Project onto PC2

$$Z_2 = X v_2$$

✅ PC2 values

$$Z_2 = \begin{bmatrix} -0.7071 \\ -0.7071 \\ 0.7071 \\ 0.7071 \end{bmatrix}$$

Step 4: Final PCA Projection Matrix

$$Z = XV = \begin{bmatrix} 1.4142 & -0.7071 \\ -1.4142 & -0.7071 \\ 1.4142 & 0.7071 \\ -1.4142 & 0.7071 \end{bmatrix}$$

🔥 Step 5: Dimensionality Reduction (IMPORTANT)

👉 If we keep only **PC1** (k = 1):

$$Z_{reduced} = \begin{bmatrix} 1.4142 \\ -1.4142 \\ 1.4142 \\ -1.4142 \end{bmatrix}$$